

PERSONAL CHARACTERISTICS AND JOB INVOLVEMENT AS ANTECEDENTS OF BOUNDARY SPANNING BEHAVIOUR: A PATH ANALYSIS

BY

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Abstract

THE relationships between age, aspects of tenure, locus of control, job involvement, and boundary spanning behaviour (B.S.B.) were examined using path analysis for 281 scientists and engineers. It was found that locus of control and age were significant determinants of job involvement. It was also shown that locus of control and job involvement were significant determinants of B.S.B. These findings are discussed relative to previous research on locus of control, job involvement, and B.S.B. Finally, new research designs are advocated which incorporate task characteristics, role dynamics constructs, and environmental uncertainty as determinants of B.S.B.

Introduction

According to Leifer and Huber¹ organizational boundary spanners are important because they function as regulators of the flow of information between the organization and its environment. Several researchers² state that boundary spanning behaviour (B.S.B.) is vital for maintaining effective organizational functioning and viability. B.S.B. is usually defined as 'the interpersonal transfer of information across organizational boundaries'.³ Since organizations are viewed as systems with permeable boundaries it becomes important to better understand the forces which influence boundary spanning behaviour. Indeed, Organ suggests that the distinctive demands made by boundary roles on position incumbents calls for a particular mix

¹ Leifer, R. and Huber, G., 'Relations among Perceived Environmental Uncertainty, Organization Structure, and Boundary Spanning Behavior', *Administrative Science Quarterly*, Vol. 22, No. 2, 1977, pp. 235-47.

² Keller, R. T., Szilagyi, A. D. and Holland, W. E., 'Boundary Spanning Activity and Employee Reactions: An Empirical Study', *Human Relations*, Vol. 29, No. 7, 1976, pp. 699-710; Aldrich, H. and Herker, D., 'Boundary Spanning Roles and Organization Structure', *Academy of Management Review*, 1977, forthcoming; Miles, R., 'Boundary Relevance: Dimensionality as a Basis for Discrepant Research Findings', *Proceedings 36th Annual Meeting of the Academy of Management*, 1976, pp. 239-43; Adams, J. S., 'The Structure and Dynamics of Behavior in Organizational Boundary Roles', in Dunnette, M. D. (Ed.), *Handbook of Industrial and Organizational Psychology*, Chicago: Rand-McNally, 1976, pp. 1175-99.

³ Keller, R. T., et al., op. cit.

of aptitudes and personality attributes.⁴ However, personality traits and biographical characteristics have not been thoroughly studied as determinants of B.S.B. Thus this study seeks to determine whether or not age, locus of control, organizational tenure, position tenure, and job involvement are antecedents of B.S.B.

The variables noted above have been shown to be interrelated by previous researchers. For instance, Oliver⁵ has shown that age is a positive antecedent of organizational tenure for his sample of 92 male life insurance agents. His path coefficient using age to predict tenure was 0.56 ($P \leq 0.01$). However, he did not investigate the relationship between age and position tenure. Most research of this type fails to assess this relationship. Other research⁶ has demonstrated a strong positive relationship between age and organizational tenure.

The personality trait locus of control⁷ has been shown to be an important determinant of organizationally valued work attitudes (job satisfaction, motivation, etc.).⁸ Locus of control is a personality trait which indicates personal beliefs about events (reinforcements) and whether they are under the control of the individual or his or her environment.⁹ Organ and Greene¹⁰ have found that locus of control was related to work satisfaction and role ambiguity. Other researchers¹¹ have found locus of control to be related to work attitudes such as job involvement and job satisfaction. Generally, as internal orientations increase, work satisfaction and job involvement increase. Lodahl and Kejner¹² have also shown that age is positively related to job involvement for a sample of nurses. Similar findings are evident in the

⁴ Hamner, W. C. and Organ, D. W., *Organizational Behavior: An Applied Psychological Approach*, Dallas: Business Publications, 1977.

⁵ Oliver, R. L., 'Antecedents of Salesmen's Compensation Perceptions: A Path Analysis Interpretation', *Journal of Applied Psychology*, Vol. 62, No. 1, 1977, pp. 20-8.

⁶ Hunt, J. W. and Saul, P. N., 'The Relationship of Age, Tenure, and Job Satisfaction in Males and Females', *Academy of Management Journal*, Vol. 18, 1975, pp. 690-702.

⁷ Rotter, J. B., 'Generalized Expectations for Internal Versus External Control of Reinforcement', *Psychological Monographs*, Vol. 80, 1966 (I, Whole No. 609).

⁸ Organ, D. W. and Greene, C., 'The Perceived Purposefulness of Job Behavior: Antecedents and Consequences', *Academy of Management Journal*, Vol. 17, No. 1, 1974, pp. 69-77; Organ, D. W. and Greene, C., 'Role Ambiguity, Locus of Control, and Work Satisfaction', *Journal of Applied Psychology*, Vol. 59, No. 1, 1974, pp. 101-102.

⁹ Rotter, J. B., op. cit.

¹⁰ Organ, D. W. and Greene, C., op. cit.

¹¹ Mitchell, T. R., Smyser, C. M. and Weed, S. C., 'Locus of Control, Supervision and Work Satisfaction', *Academy of Management Journal*, Vol. 18, 1975, pp. 623-31; Pryer, M. W. and Distefano, Jr., M. K., 'Perceptions of Leadership Behavior, Job Satisfaction, and Internal-External Control Across Three Nursing Levels', *Nursing Research*, Vol. 20, 1971, pp. 534-7; Runyon, K. E., 'Some Interactions Between Personality Variables and Management Styles', *Journal of Applied Psychology*, Vol. 57, No. 3, 1973, pp. 288-94.

¹² Lodahl, T. M. and Kejner, M., 'The Definition and Measurement of Job Involvement', *Journal of Applied Psychology*, Vol. 49, No. 1, 1965, pp. 24-33.

research of Dewhirst and Avery in their sample of scientists and engineers.¹³ However, neither of the previous two research studies controlled for the effects of tenure or locus of control on the age-job involvement relationship.

McKelvey and Sekaran¹⁴ note that job involvement is particularly important in research organizations where professionals are employed and quality of their performance and amount of their output is especially sensitive to intrinsic rewards, levels of satisfaction and commitment to job performance. However, only limited research has been conducted which assesses the relationship between job involvement, locus of control, and B.S.B.

As noted earlier, B.S.B. is generally defined by researchers as the interpersonal transfer of information across organizational boundaries or communication frequency with external agents.¹⁵ McKelvey and Sekaran¹⁶ have shown that job involvement is significantly related to an aspect of B.S.B., namely interest in innovation ($r = 0.28$, $P \leq 0.0005$). Further, Ruble¹⁷ notes that a variety of studies¹⁸ have shown that individuals who view their outcomes as determined by themselves (internals) exhibit more initiative in their attempts to control their environment than individuals who view their outcomes as determined by outside forces (externals). Certainly one behavioural response for internals who are involved with their jobs would be boundary spanning behaviour. Also, research conducted by Lodahl and Kejner,¹⁹ has shown that perceived job involvement is positively related to B.S.B. ($r = 0.30$, $P \leq 0.01$) as measured by the number of people contacted *per day* by respondents in their sample of 70 engineers.

The research presented here will specifically attempt to test several lines of thought developed in the literature review presented above. Age, locus of control, organizational tenure and position tenure are all hypothesized to be antecedents of job involvement. Secondly, it is hypothesized that locus of control is unrelated to age and facets of tenure; but that it is a negative antecedent of job involvement and B.S.B. Finally, it is hypothesized that job involvement will moderate the relationships between age, tenure facets, and

¹³ Dewhirst, D. and Avery, R., 'Range of Interests Versus Job Performance and Satisfaction', *Research Management*, Vol. 19, 1976, pp. 18-23.

¹⁴ McKelvey, B. and Sekaran, U., 'Toward a Career-Based Theory of Job Involvement: A Study of Scientists and Engineers', *Administrative Science Quarterly*, Vol. 22, No. 2, 1977, pp. 281-305.

¹⁵ Keller, R. T., Szilagyi, A. D. and Holland, W. E., *op. cit.*

¹⁶ McKelvey, B. and Sekaran, U., *op. cit.*

¹⁷ Ruble, T. L., 'Effects of One's Locus of Control and the Opportunity to Participate in Planning', *Organizational Behavior and Human Performance*, Vol. 16, No. 1, 1976, pp. 63-73.

¹⁸ Lefcourt, H. M., 'Internal Versus External Control of Reinforcement: A Review', *Psychological Bulletin*, Vol. 65, No. 4, 1966, pp. 206-220; Phares, E. J., Ritchie, D. E. and Davis, W. L., 'Internal-External Control and Reaction to Threat', *Journal of Personality and Social Psychology*, Vol. 10, No. 4, 1968, pp. 402-5.

¹⁹ Lodahl, T. M. and Kejner, M., *op. cit.*

B.S.B. All of the hypothesized relationships detailed above are presented in figure 1 and table I below. In table I a zero in place of a beta coefficient indicates that no relationship is hypothesized between that particular antecedent variable and the given criterion. Finally, it should be noted that hypothesized significant coefficients for locus of control are negative. Thus, as subjects increase in internality (represented by smaller scores), they should be more involved in their jobs and more oriented towards the transfer of information across organizational boundaries.

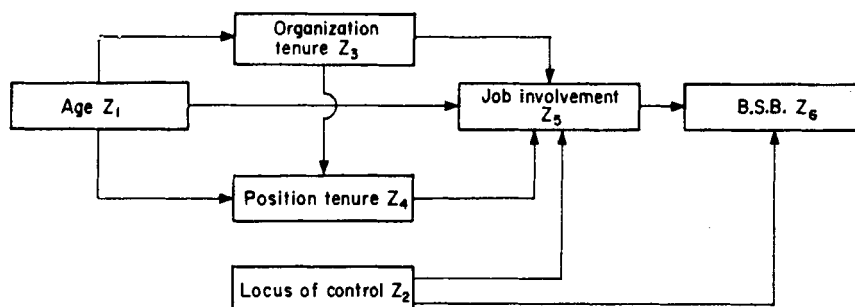


Fig. 1. Hypothesized relationships between age, facets of tenure, job involvement, and B.S.B.

Table I. Hypothesized relationships between B.S.B. and antecedent variables

Variable		Structural equations
Age:	Z_1	—
Locus of control:	Z_2	$OZ_1 + e_2$
Organization tenure:	Z_3	$b_{31}Z_1^* + OZ_3 + e_3$
Position tenure:	Z_4	$b_{41}Z_1^* + OZ_4 + b_{43}Z_3^* + e_4$
Job involvement:	Z_5	$b_{51}Z_1^* - b_{52}Z_2^* + b_{53}Z_3^* + b_{54}Z_4^* + e_5$
B.S.B.:	Z_6	$OZ_1 - b_{62}Z_2^* + OZ_3 + OZ_4 + b_{65}Z_5^* + e_6$

* A significant coefficient is hypothesized.

Method

Sample. The study was conducted in 15 public and private research organizations located in the Western United States. Two hundred and eighty-one scientists and engineers participated in the study. Forty-five of the respondents were project leaders while the remainder were team members. Since participation in the study was voluntary, the 281 questionnaires returned did not represent all of the technical people employed by the 15 organizations. Total employment of scientists and engineers was 961, therefore the actual response rate was 29.3 per cent. These organizations ranged in size from five to 450 technical people employed. Organizational response rates were six with 100 per cent, one above 80 per cent, four between 50 and

80 per cent, two between 20 and 50 per cent, and two less than 20 per cent. Generally, larger organizations provided the most respondents but the lowest response rates. Questionnaires were distributed personally by the researcher and anonymity was guaranteed to participants and organizations.

The average age of respondents was 38.8 (SD = 8.1) years, and the average time with the organization was 8.5 (SD = 6.7) years. The average time in position was 6.3 (SD = 6.6) years. Average job involvement was 4.2 (SD = 0.94), and B.S.B. averaged 14.4 (SD = 6.0). Means for the four items of the B.S.B. scale were 3.3, 4.3, 4.0, and 2.8 respectively. Thus none of the items in the scale appeared to be heavily influencing the composite mean. Ninety-eight percent of the respondents were college graduates, 26 per cent held Ph.D. degrees, and 18 per cent held Master's degrees.

Technical activities for members of the participating organizations were largely confined to basic research and the elaboration of technological systems designed for space exploration. Thus little applied research was evidenced in this sample. The majority of the project groups and organizations were involved in research projects funded by agencies of the U.S. federal government. Specifically, these projects dealt with the environmental sciences (climatology, geological, weather, and solar research), high energy physics, and the development of instrumentation for deep space exploration.

Measures. Boundary Spanning Behaviour (B.S.B.) was measured with a four-item instrument developed by Keller, et al.²⁰ The questions in the instrument deal with obtaining information from sources outside the organization (both written and fellow professionals). Validity and reliability estimates for the instrument are presented in Keller, et al.²¹ Job involvement was measured by one item which has been reported as being valid and reliable in other research settings.²² The question takes the following form:

Some individuals are completely involved in their technical work — absorbed by it night and day. For others, their work is simply one of several interests. How involved do you feel in your work. CHECK ONE.

Not much involved; Slightly involved, Moderately; Strongly; Very strongly; Completely absorbing.

Single item measures are frequently criticized for their lower reliability, the result of random measurement error which lowers the magnitude of the tested statistical relationship.²³ However, if significant relationships between

²⁰ Keller, R. T., et al., op. cit.

²¹ Ibid.

²² Patchen, M., *Participation, Achievement and Involvement on the Job*, Englewood Cliffs, N.J.: Prentice-Hall, 1970.

²³ Blalock, H. M., *Social Statistics*, New York: McGraw-Hill, 1972.

dependent and independent variables measured with single predictors are uncovered, these relationships are usually reliably sound and they would be larger if the measurement of the variables were more reliable.

The personality construct locus of control was measured by using Valencha's²⁴ shortened version of Rotter's²⁵ instrument. The scale measures personal beliefs about events (reinforcements) and whether they are under the control of the individual or his or her environment. Internal orientations are indicated by smaller while external orientations are indicated by larger scores.

Tenure was measured in two ways in this research. Respondents were asked to indicate to the nearest month how long they had worked in the organization (organizational tenure) and, in the same format, how long they had occupied their present position (position tenure). These results were verified independently by examining organizational records for the personnel decisions of hiring and promotion. Age was measured by asking respondents how old they were in years.

Analysis. Path analysis was employed to test for the relationships suggested in the prior discussion, in table I, and in figure 1. Descriptions of this technique are presented in Blalock and Blalock;²⁶ and Blalock.²⁷ Basically, the existence of an hypothesized causal linkage is represented by the significance of the standardized beta weight or path coefficient generated when a variable in the theoretical model is regressed on all variables hypothesized to be antecedent to it. Thus, in the absence of a direct effect between variables, the corresponding path coefficient should not be significant.²⁸

Results

Table II presents the zero-order correlations among the study variables. From the table it can be seen that both of the antecedents of position tenure were significant positive correlates. Second, it can be seen that both age and locus of control are significant correlates of job involvement. Finally, age, locus of control and job involvement are all significantly related to B.S.B.

Table III shows the results generated when the hypothesized antecedents were entered into the path analysis in the order of hypothesized precedence. The data reveal that thirteen of fifteen hypothesized relationships were

²⁴ Valencha, G. K., 'Construct Validation of Internal-External Locus of Control of Reinforcement Related to Work Related Variables', *Proceedings of the American Psychological Association*, 1972, pp. 455-6.

²⁵ Rotter, J. B., op. cit.

²⁶ Blalock, H. M. and Blalock, A., *Methodology in Social Research*, New York: McGraw-Hill, 1968.

²⁷ Blalock, H. M., *An Introduction to Social Research*, Englewood Cliffs, N.J.: Prentice-Hall, 1970.

²⁸ Oliver, R. L., op. cit.

Table II. Zero-order correlations among study variables

	(1)	(3)	(4)	(5)	(6)
<i>Age</i>	(1) 0.430**	0.286**	0.020	0.102*	0.090
<i>Organizational tenure</i>	(2)	0.392**	-0.030	0.038	0.046
<i>Position tenure</i>	(3)		0.037	-0.048	-0.013
<i>Locus of control</i>	(4)			-0.329**	-0.233**
<i>Job involvement</i>	(5)				0.333**
<i>B.S.B.</i>	(6)				—

** $P \leq 0.001$ * $P \leq 0.05$

supported. This is true since table I hypothesized that fifteen coefficients existed as being either positive, negative, or zero. More specifically, table III indicates that both age and organizational tenure are significant antecedents of position tenure. Also, the results confirm that age and locus of control are antecedents of job involvement. The negative coefficient for locus of control in the regression equation predicting job involvement ($B = -0.329$, $P \leq 0.001$) indicates that internals are more involved with their jobs than externally oriented scientists. It should be recalled that smaller locus of control scores indicate increasing internal orientations. Neither organizational tenure nor position tenure demonstrated the hypothesized positive path coefficients with job involvement. Indeed, organizational tenure possessed a slightly negative path coefficient.

Table III. Path coefficients corresponding to antecedents of B.S.B.

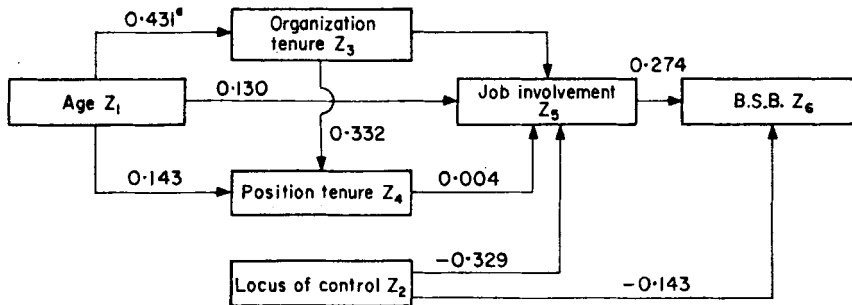
<i>Variable</i>	<i>Structural equations</i>	<i>R²</i>
<i>Age</i>	Z_1	—
<i>Locus of control</i>	Z_2 0.020 Z_1	0.0004
<i>Organizational tenure</i>	Z_3 0.431 Z_1^* - 0.039 Z_2	0.1867**
<i>Position tenure</i>	Z_4 0.143 Z_1^* + 0.044 Z_2 + 0.332 Z_3^*	0.1727**
<i>Job involvement</i>	Z_5 0.130 Z_1^* - 0.329 Z_2^* - 0.074 Z_3 + 0.004 Z_4	0.1242**
<i>B.S.B.</i>	Z_6 0.065 Z_1 - 0.143 Z_2^* + 0.010 Z_3 - 0.010 Z_4 + 0.274 Z_5^*	0.1300**

Note: Values in structural equations are standardized path coefficients.

** $p \leq 0.001$.

For the study's criterion B.S.B. all of the hypothesized path coefficients were supported. Thus only locus of control and job involvement were related to B.S.B. It should be noted that job involvement was more strongly related to B.S.B. than locus of control as the model in figure 1 would hypothesize. The regression model for B.S.B. explained 13 per cent of the variance in the criterion ($F = 8.21$, $P \leq 0.001$). As is generally true for path analysis, predictive power increases as the number of hypothesized antecedents decreases. Thus, the best predictive power occurs for organizational tenure ($R^2 = 0.1867$; $F = 31.9$, $P \leq 0.001$).

All of the findings in table III are indicated in figure 2. From the figure it can be seen that the original hypothesized model held up very well indeed. The only discrepancy between the empirical model in figure 2 and the theoretical model in figure 1 occurred with respect to the non-significant tenure antecedents for job involvement. All other relationships were supported.



* Standardized Beta coefficients (path coefficients).
Fig. 2. Path diagram of the antecedents of B.S.B. suggested by the study results

Discussion

The results of the study indicate that several potential determinants of job involvement and boundary spanning behaviour have been identified in this sample of scientists and engineers. Organizational tenure was found to be partially determined by age, whilst position tenure was shown to be determined by age and organizational tenure. The results confirm and extend the research of Oliver,²⁹ Organ and Green³⁰ and Hunt and Saul.³¹

The present study indicated that age and facets of tenure were not strong determinants of job involvement or B.S.B. To the extent that B.S.B. is an important position requirement for scientists and engineers, R and D managers should place greater emphasis on locus of control and job involvement in their human resource decisions. Indeed, an internal orientation coupled with high levels of job involvement should be factors that are considered for incumbents of any boundary spanning role. The presence of these two characteristics should greatly enhance the boundary role spanner's ability to cope with uncertainty and conflict inherent in the position.

The findings of the path analysis also indicated that locus of control was a strong negative antecedent of job involvement, whilst age was a positive antecedent. The locus of control finding confirms and extends the research

²⁹ Oliver, R. L., op. cit.

³⁰ Organ, D. W. and Greene, C., op. cit.

³¹ Hunt, J. W. and Saul, P. N., op. cit.

of Mitchell, et al.,³² Pryer and Distefano,³³ and Runyon.³⁴ All of these researchers found locus of control to be negatively related to job involvement or job satisfaction. These results tend to suggest that older more internally oriented professionals are going to be more involved with their jobs. This interpretation is consistent with research conducted by Dewhirst and Avery³⁵ on scientists and engineers. These researchers also found that job involvement moderated the relationship between age and performance.

Given the above findings, R and D managers may wish to consider the validity of creating project groups where some of the members are older scientists with internal orientations. These individuals would be more likely to exhibit high job involvement and act as resources for younger and less self-confident (externally oriented) project members. Indeed, such human resource practices would probably lead to more cohesive and collaborative project groups.

Neither organizational nor position tenure were positive antecedents of job involvement. One reason for this finding could be that task and team characteristics are more important for scientists and engineers. Research in the R and D as well as in other settings have shown that task and group characteristics are related to job involvement.³⁶ Important task characteristics include perceived uncertainty and structure; while salient group variables are cohesiveness, size, and collaboration. It would therefore be useful to expand future models of job involvement to include some of these variables as predictors or antecedents.

The study presented here indicated that both job involvement and locus of control were antecedents of B.S.B. Thus, as scientists and engineers became simultaneously more involved in their work and more internal in their orientations, they were more likely to exhibit B.S.B. Job involvement and internal locus of control may be antecedents of other valued outcomes in R and D settings such as creativity and invention. Additional research needs to be performed to adequately detail such relationships.

Even though job involvement and locus of control were found to be antecedents of B.S.B. they explained only 13 per cent of the variance in the criterion. One reason for this may have been the nature of the locus of control construct. Current research has indicated that several dimensions of the construct exist such as easiness, justness, predictability, and political

³² Mitchell, T. R., et al., op. cit.

³³ Pryer, M. W. and Distefano, Jr., M. K., op. cit.

³⁴ Runyon, K. E., op. cit.

³⁵ Dewhirst, D. and Avery, R., op. cit.

³⁶ Cartwright, D., 'The Nature of Group Cohesiveness', in Cartwright, D. and Zander, A. (Eds.), *Group Dynamics*, New York: Harper and Row, 1969, pp. 91-109; Hackman, R. J. and Vidmar, N., 'Effects of Size and Task Type on Group Performance and Member Reactions', *Sociometry*, Vol. 33, No. 1, 1970, pp. 33-45.

responsiveness of the world.³⁷ Additional research must be performed before variation in B.S.B. can be completely understood relative to these emerging facets of locus of control.

It would be useful to redesign the locus of control instrument so that it more accurately reflects internal and external orientations in work organizations. Currently, the instrument reflects an orientation towards the origin of reward contingencies in general life patterns. A narrowing of the scope of the instrument to work organizations could be very beneficial for future research on the construct.

It is also apparent from other research³⁸ that additional variables besides job involvement and locus of control must be included in models predicting B.S.B. Generally, these variables would include role dynamics constructs (ambiguity and conflict), task characteristics (uncertainty, complexity, and autonomy) and environmental uncertainty. Future research designs should incorporate these variables in a multivariate framework for improving an understanding of boundary spanning behaviour.

³⁷ Duffy, P. J., Shiflett, S. and Downey, R. G., 'Locus of Control: Dimensionality and Predictability Using Likert Scales', *Journal of Applied Psychology*, Vol. 62, No. 2, 1977, pp. 214-9; Collins, B. E., 'Four Components of the Rotter Internal-External Scale: Belief in a Difficult World, a Just World, a Predictable World, and a Politically Responsive World', *Journal of Personality and Social Psychology*, Vol. 29, No. 3, 1974, pp. 381-91.

³⁸ Leifer, R. and Huber, G., op. cit.; Aldrich, H. and Herker, D., op. cit.; Keller, R. T. et al., op. cit.; Miles, R., op. cit.